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QTJ

QUARTERLY EXAMINATION - 2026

12 - Std CHEMISTRY

Time : 3.00 hrs. Marks : 70

I. Answer All the Questions 15 x 1 = 15

1. Wolframite ore is separated from tinstone by the process of
   a) Smelting b) Calcination c) Roasting d) Electromagnetic separation
2. Silicones form silicon resins with: a) Acrylic esters b) Alkanes c) Methylene d) Chlorides
3. The compound that is used in nuclear reactors as protective shields and control rods is
   a) Metal borides b) Metal oxides c) Metal carbonates d) Metal carbides
4. **Assertion:** Bond dissociation energy of fluorine is greater than chlorine gas
   Reason: Chlorine has more electronic repulsion than fluorine
   a) Both assertion and reason are true and reason is the correct explanation of assertion.
   b) Both assertion and reason are true but reason is not the correct explanation of assertion.
   c) Assertion is true but reason is false. d) Both assertion and reason are false.
5. Which is used in the dyeing and printing industry? a) $K_2Cr_2O_7$ b) $KMnO_4$ c) CrO_2Cl_2 d) Na_2CrO_4
6. The actinoid elements which show the highest oxidation state of +7 are
   a) Np, Pu, Am b) U, Fm, Th c) U, Th, Md d) Es, No, Lr
7. The vacant space in bcc lattice unit cell is a) 48% b) 23% c) 32% d) 26%
8. The yellow colour in NaCl crystal is due to
   a) excitation of electrons in F centers b) refraction of light from Na^+ ion
   c) reflection of light from Cl^- ion on the surface d) all of the above
9. What does a catalyst change during a reaction?
   a) Enthalpy b) Activation energy c) Entropy d) Internal energy
10. Which of these is not likely to act as Lewis base? a) BF_3 b) PF_3 c) CO d) F^-
11. The pH of an aqueous solution is Zero. The solution is
    a) slightly acidic b) strongly acidic c) neutral d) basic
12. Oxidation of glycerol with bismuth nitrate produces:
    a) Meso-oxalic acid b) Tartronic acid c) Formaldehyde d) Glyceraldehyde
13. The reagent used to distinguish between acetaldehyde and benzaldehyde is
    a) Tollens reagent b) Fehling's solution c) 2,4 – dinitrophenyl hydrazine d) semicarbazide
14. Phenyl-methanal is reacted with concentrated NaOH to give two products X and Y. X reacts with metallic sodium to liberate hydrogen. X and Y are
    a) Sodium benzoate and phenol b) Sodium benzoate and phenyl methanol
    c) Phenyl methanol and sodium benzoate d) none of these
15. In which of the following reactions new carbon – carbon bond is not formed?
    a) Aldol condensation b) Friedel craft reaction c) Kolbe's reaction d) Wolf kishner reduction

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II. Answer Any Six Questions (Question No.24 is Compulsory) 6 x 2 = 12

16. Distinguish between Mineral and Ore.
17. Write the uses of Carbon dioxide.
18. Write a short note on Frenkel defect.
19. Complete the reaction: $CH_3OH + SOCl_2 \rightarrow$
20. What are Interhalogen compounds? Give an example.
21. Which is more stable, Fe^{3+} or Fe^{2+}? Why?
22. Explain Pseudo first order reaction with an example.
23. Write the Williamson ether synthesis method.
24. Identify the Lewis acid and Lewis base in: $Cr^{3+} + 6H_2O \rightarrow [Cr(H_2O)_6]^{3+}$

III. Answer Any Six Questions (Question No.33 is Compulsory) 6 x 3 = 18

25. What are Interstitial compounds?
26. Explain the Alumino-thermic process.
27. Write the uses of Ethylene glycol.
28. How does ionization enthalpy vary from Aluminium to Thallium?
29. Write the coordination number of AAAA, ABABA and ABCABC type of three-dimensional packing.
30. Write the Hell-Volhard-Zelinsky (HVZ) reaction.
31. For a first order reaction A → Product, 60% of A decomposes in 40 minutes. Calculate the half-life.
32. Define Solubility product.
33. What type of hybridization is found in the following i) BrF_3 ii) BrF_5 iii) BrF

IV. Answer All the Questions. 5 x 5 = 25

34. (a) Explain Zone refining method with an example. **OR**

(b) (i) What is Producer gas?

(ii) Complete: A. $B(OH)_3 + NH_3 \rightarrow ?$ B. $Na_2B_4O_7 + H_2SO_4 + H_2O \rightarrow ?$ C. $SiCl_4 + NH_3 \rightarrow ?$

35. (a) (i) Describe the structure of Ammonia. (ii) Write the thermal reaction of Phosphorus acid.

OR

(b) (i) Write the example for oxidising properties of Potassium dichromate.

(ii) Actinoid contraction is greater than Lanthanoid contraction. Why?

36. (a) Explain the collision theory for bimolecular reactions.

OR

(b) (i) Differentiate between octahedral and tetrahedral voids.

(ii) Classify the following solids. a. P_4 b. Brass c. Sodium chloride d. I_2

37. (a) Derive the Henderson-Hasselbalch equation.

OR

(b) (i) Define polar molecular crystals. (ii) Explain Acid leaching and Cyanide leaching.

38. (a) How will you prepare phenol from: (i) Chlorobenzene (ii) Isopropylbenzene.

OR

(b) C_7H_6O (A) reacts with aldehyde in presence of base to give (B), and (A) reacts with carboxylic acid in presence of pyridine to eliminate CO_2 to give (C). Identify A, B and C and write the reactions.

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